

Practical Application of Regularization Techniques in ML

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[Next item →](#)[Instructions](#)

1. In the context of a neural network model trained on a complex dataset, you observe that the model performs well on training data but poorly on validation data. What might this indicate about the role of regularization in your model? 1 / 1 point
- ☒ The model might be overfitting the training data, suggesting that regularization could help improve performance on the validation set.
 - ☐ The model has a balanced bias-variance trade-off, and regularization would have no effect.
 - ☐ The model is underfitting, and regularization would further increase the bias.
 - ☐ The model's training accuracy is too high, suggesting regularization is needed to reduce it.
- ☒ **Correct**
Correct! Overfitting occurs when the model learns the training data too well, and regularization can help improve generalization to the validation set.
2. Consider a neural network with L1 regularization applied. How would you expect the regularization to impact the model's weights and overall complexity? 1 / 1 point
- ☐ L1 regularization uniformly scales down all weights without affecting their sparsity.
 - ☐ L1 regularization makes the model more complex by adding additional parameters.
 - ☒ L1 regularization tends to drive some weights to zero, resulting in a sparse model that can be simpler and more interpretable.
 - ☐ L1 regularization increases the weights' magnitude to avoid overfitting.
- ☒ **Correct**
Correct! L1 regularization promotes sparsity in the weights, which can simplify the model and make it more interpretable.
3. You are tuning a neural network and find that applying dropout consistently improves the model's performance on the validation set. Which of the following best explains why dropout is beneficial in this scenario? 1 / 1 point
- ☒ Dropout reduces overfitting by preventing the network from becoming too reliant on any single neuron during training.
 - ☐ Dropout adds noise to the data, which is unrelated to model generalization.
 - ☐ Dropout increases the training error by reducing the network's capacity.
 - ☐ Dropout speeds up training by reducing the amount of data processed per epoch.
- ☒ **Correct**
Correct! Dropout helps to generalize better by randomly ignoring neurons during training, which reduces overfitting.
4. When comparing models with and without L2 regularization on a dataset with significant noise, what changes might you observe in the decision boundaries of the models? 1 / 1 point
- ☒ The decision boundaries of the model with L2 regularization are likely to be smoother and less sensitive to the noise compared to the model without regularization.
 - ☐ The decision boundaries without L2 regularization will always be smoother than those with regularization.
 - ☐ Both models will have identical decision boundaries regardless of regularization.
 - ☐ The model with L2 regularization will have more complex decision boundaries due to increased regularization strength.
- ☒ **Correct**
Correct! L2 regularization helps to smooth decision boundaries and reduces the sensitivity to noise.
5. In a scenario where a neural network is trained on a large, high-dimensional dataset, you notice that adding L1 regularization improves the model's performance by increasing sparsity in the weights. What is the practical implication of this observation for the model? 1 / 1 point
- ☐ The model's performance is negatively affected due to an increase in the number of non-zero weights.
 - ☒ The model becomes more interpretable and computationally efficient due to reduced dimensionality and fewer non-zero weights.
 - ☐ The model becomes less interpretable due to the addition of more features.
 - ☐ The model's complexity increases due to the introduction of additional parameters.
- ☒ **Correct**
Correct! L1 regularization helps in reducing dimensionality and making the model more efficient and interpretable.